Graph Neural Networks for Protein Folding

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***Abstract*—Proteins fold into shapes that decide what they do. We treat a protein as a graph of its residues and learn its structure by passing messages along the graph. On a standard benchmark the model predicts contact maps more accurately than sequence-only baselines.**

I. Introduction

Predicting how a protein folds from its sequence is a central problem in structural biology [1]. Each residue interacts with the residues it touches, which makes a graph a natural description of the structure [2].

II. Method

A graph neural network updates the state of each residue from the states of its neighbours:

$$h_v^{(k+1)} = \sigma\Big(W^{(k)} \sum_{u \in N(v)} h_u^{(k)}\Big) \tag{1}$$

Fig. 1 shows the architecture.

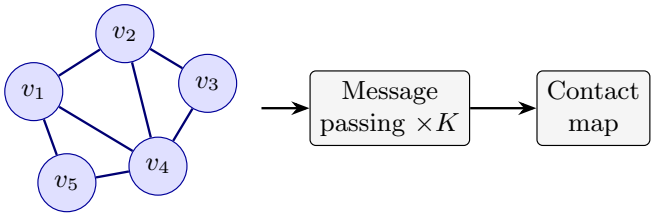


Fig. 1. The model: residues are nodes and contacts are edges.

III. Results

Table I compares the model with two baselines from [3]. The model improves the Top-L precision by 0.09.

TABLE I
Precision of contact prediction

Model	Top-L	Top-$L/5$
Sequence CNN	0.41	0.63
Transformer	0.52	0.74
Ours	**0.61**	**0.82**

References

[1] J. Jumper, R. Evans, A. Pritzel *et al.*, "Highly accurate protein structure prediction with AlphaFold," *Nature*, vol. 596, pp. 583–589, 2021.

[2] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1263–1272.

[3] R. Rao, J. Meier, T. Sercu, S. Ovchinnikov, and A. Rives, "Transformer protein language models are unsupervised structure learners," in *International Conference on Learning Representations*, 2021.